SOME TECHNICAL NOTES

The laws of physics that govern our universe are expressed in the language of mathematics. For readers comfortable with math, I write down a few formulas that come from the physical laws and show how I used them to deduce some things in this book. Two numbers that appear frequently in my formulas are the speed of light, $c = 3.00 \times 10^{8}$ meters/second, and Newton's gravitational constant, $G = 6.67 \times 10^{-11}$ meters^3/kilogram/second^2. I use scientific notation so 10^{8} means 1 with eight zeros after it, 100,000,000 or a hundred million, and 10^{-11} means 0.[ten zeros]1, that is, 0.00000000001. I don't aspire to accuracy any higher than 1 percent, so I show only two or three digits in my numbers, and when a number is very poorly known, only one digit.

Chapter 4. Warped Time and Space, and Tidal Gravity

The simplest, quantitative form of Einstein's law of time warps is this: Place two identical clocks near each other, and at rest with respect to each other, separated from each other along the direction of the gravitational pull that they feel. Denote by R the fractional difference in their ticking rates, by D the distance between them, and by g the